Renfeng Peng

renfpeng (a.t.) cityu.edu.hk
http://jimmypeng1998.github.io/
Sept. 30th, 1998, Changde, Hunan Renfeng Peng at Google scholar
Workstation 7, Room 215, Innovation Centre, 72 Tat Chee Avenue, Kowloon Tong, Kowloon, Hong Kong



Employment

2026.07 – Present **Research Assistant**
Department of Data Science, City University of Hong Kong, Hong Kong SAR
Host: Prof. Fenglei Fan

Education

2021.09 – 2026.06 **Ph.D. in computational mathematics**
Academy of Mathematics and Systems Science, Chinese Academy of Sciences; and University of Chinese Academy of Sciences, Beijing, China
Supervisor: Prof. Ya-xiang Yuan

2017.09 – 2021.07 **Bachelor in mathematics and applied mathematics**
School of Mathematical Sciences, Tongji University, Shanghai, China.

Publications

Refereed papers

1. Zhengyang Liu, Jianguo Huang, Junxi Hu, Yucheng Mi, Haojie Cai, Wenxuan Shen, Zhikun Zhang, **Renfeng Peng**, Guangtao Zhang, and Zhiqiang Liu. "Physics-Informed Functional Tucker Method with RKHS Factors for Sparse Spatiotemporal Reconstruction". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Accepted. 2026. URL: https://arxiv.org/abs/2611.03717.
2. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. "Low-rank optimization on Tucker tensor varieties". In: *Mathematical Programming* 214 (2025), pp. 357–407. DOI: 10.1007/s10107-024-02186-w.
3. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. "Optimization on product manifolds under a preconditioned metric". In: *SIAM Journal on Matrix Analysis and Applications* 46.3 (2025), pp. 1816–1845. DOI: https://epubs.siam.org/doi/abs/10.1137/24M1643773.
4. Chengkai Zhu, **Renfeng Peng**, Bin Gao, and Xin Wang. "Riemannian Optimization for Holevo Capacity". In: *2025 IEEE International Symposium on Information Theory (ISIT)*. 2025, pp. 1–6. DOI: 10.1109/ISIT63088.2025.11195337.
5. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. "Riemannian preconditioned algorithms for tensor completion via tensor ring decomposition". In: *Computational Optimization and Applications* 88 (2024), pp. 443–468. DOI: 10.1007/s10589-024-00559-7.
6. Ling-Xi Zhang, **Ren-Feng Peng**, and Jun-Feng Yin. "A second order numerical scheme for fractional option pricing models". In: *East Asian J. Appl. Math* 11 (2021), pp. 326–348. DOI: 10.4208/eajam.020820.121120.

Preprints

1. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Quotient geometry of tensor ring decomposition”. In: *ArXiv preprint arXiv:2601.21874* (2026).
2. Fan Gao, Wei Su, Juntong Fan, **Renfeng Peng**, Hongyu Liu, Jinqiao Duan, and Feng-Lei Fan. “Dynamics as Code: On Model Compression via Dynamic System”. In: *ArXiv preprint arXiv:2610.11115* (2026).
3. Pengfei Hao, **Renfeng Peng**, Yang Kuang, and Bin Gao. “High-dimensional extreme eigenvalue problems: low-rank tensor parametrization and optimization”. In: *ArXiv preprint arXiv:2609.30138* (2026).
4. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “First-order methods on bounded-rank tensors converging to stationary points”. In: *ArXiv preprint arXiv:2503.04523* (2025).
5. **Renfeng Peng**, Chengkai Zhu, Bin Gao, Xin Wang, and Ya-xiang Yuan. “Normalized tensor train decomposition”. In: *ArXiv preprint arXiv:2511.04369* (2025).
6. Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Desingularization of bounded-rank tensor sets”. In: *ArXiv preprint arXiv:2411.14093* (2024).

Research Interests and Skills

Research	Tensor optimization, manifold optimization, and machine learning.
Languages	Mandarin Chinese, English.
Coding	Matlab, C/C++, LaTeX, ...

Awards and achievements

2026/07	**CAS President’s Excellent Award (5,000 CNY).**
2025/09	**AMSS President’s Special Award (32,000 CNY).**
2024/09	**Loo-Keng Hua Scholarship.**
2021/09	**Loo-Keng Hua Scholarship.**

Presentations

2026/04	**2026 International Conference on Optimization: Techniques and Applications (ICOTA 2026).** Shanghai, China.
2025/08	**The 15th International Conference on Numerical Optimization and Numerical Linear Algebra.** Harbin, China.
2025/07	**The 8th International Conference on Continuous Optimization.** Los Angeles, U.S.
2024/11	**The 7th PKU PhD Symposium for Applied Mathematics.** Beijing, China.

Referee for journals

- Mathematical Programming Computation
- Optimization Methods and Software